Fuad Hasan

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Research Profile

Computer Science and Engineering graduate with three peer-reviewed conference publications, including two as first author. Research spans explainable threat detection, Android malware detection and intrusion detection in Internet of Medical Things (IoMT) networks, with undergraduate thesis work in Bengali natural language processing.

Research Interests

Machine learning for cybersecurity; explainable AI; IoT and IoMT security; network intrusion and Android malware detection; feature selection and ensemble learning.

Education

Bachelor of Science in Computer Science and Engineering July 2026
Sylhet International University, Sylhet, Bangladesh
CGPA: 3.61/4.00

Peer Reviewed Conference Publications

1. **Fuad Hasan**, Priyanka Roy, and Ashraful Goni. "Towards a XAI-Enhanced Machine Learning Framework for Unmasking Cybersecurity Threats with Intelligent Feature Scoring in Telesurgery." *2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence and Networking (QPAIN), 2026.*
   DOI: 10.1109/QPAIN69676.2026.11546513
2. **Fuad Hasan**, Ashraful Goni, Farjana Akter, and Md Umor Faruk Jahangir. "BNFF-WaveGraph: A BiDirectional Node Fusion-Fission Hybrid Deep Learning Architecture for Android Malware Detection." *2025 IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE), pp. 1267-1272, 2025.*
   DOI: 10.1109/WIECON-ECE69386.2025.11526121
3. Rubiat Jahan Momtahina, Abdullah Rajib, Ashraful Goni, and **Fuad Hasan**. "Genetic Algorithm with Optimised Extra Tree Ensemble for Enhanced Intrusion Detection in IoMT Networks." *2025 IEEE International WIE Conference on Electrical and Computer Engineering (WIECON-ECE), pp. 1273-1278, 2025.*
   DOI: 10.1109/WIECON-ECE69386.2025.11526080

Thesis Manuscript Under Review

MCBA-Net: A Hybrid Deep Architecture Framework for Bengali News Classification.

Undergraduate thesis manuscript. Under review at Computer Science (AGH University of Krakow).

Studied deep learning for Bengali news classification, including violence-related news categories.

Selected Research Projects

Explainable threat detection for telesurgery 2026

• Investigated machine learning for threat detection and feature scoring. The study evaluated a Kaggle telesurgery dataset and Edge-IIoTSet with SHAP-based explanations.

Android malware detection with hybrid deep learning 2025

• Co-authored a study of a bidirectional node fusion-fission hybrid deep learning architecture for Android malware detection.

IoMT intrusion detection with optimised Extra Trees 2025

• Co-authored a study combining genetic algorithm optimisation with an Extra Trees ensemble for intrusion detection in IoMT networks.

Thyroid cancer prediction using machine learning 2024

• Prepared data, trained models and evaluated thyroid cancer prediction using Python, Pandas, NumPy, Scikit-learn and Matplotlib.

Technical Skills

Programming: Python, C++, Java, JavaScript
Machine learning: Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, Matplotlib
Tools and databases: Git, GitHub, Google Colab, VS Code, MySQL
Web development: HTML, CSS, Bootstrap, Liquid, Shopify theme development

Professional Experience

Junior Shopify Developer, ArtifyLab Oct 2025 - Present

Remote | Customise Shopify themes using Liquid, JavaScript, HTML and CSS; implement features and troubleshoot issues with a remote team.

Exam Invigilator / Exam Marshal, British Council Nov 2023 - Present

Sylhet, on-call | Support IELTS and Cambridge examination administration, security and candidate verification.

Technical Content Specialist, 10 Minute School Oct 2018 - Sep 2025

Remote | Prepared technical and educational content and supporting learning materials for students.

Awards and Leadership

• 1st Runner-up, SIU ICT Fest 2025 Intra-University Programming Contest.
• Coder of the Month, Department of CSE, Sylhet International University, February 2025.
• ICPC Dhaka Regional onsite participant, 2023; inter-university programming contest participant, 2022-2025.
• Solved 500+ programming problems across online judges. Codeforces | LeetCode
• Assistant General Secretary, SIU CSE Society, May 2023 - June 2025; supported student activities and academic events.
• Volunteer, Bangladesh Physics Olympiad, Sylhet Regional, 2024-2026; Campus Ambassador, FutureNation (UNDP) and bdapps.

Academic References

Md Umor Faruk Jahangir | Assistant Professor and Head
Department of CSE, Sylhet International University | jahangircs@siu.edu.bd

Khaled Hussain | Dean and Associate Professor
Department of CSE, Sylhet International University | khussain@siu.edu.bd